

BT 503 Mid Current Solved

MCQ's

- Which of does not use to make vessel of biofilters (may be formic sheet)
- Which of the following is not fate of water (may be generation of aerosol)
- Which of the following does not classify as a substitution material?
- Which of the following statement about desulphurization of oil and coal is false?
- Metals are released into the environment by all of the following except (Earthquake)
- When facultative pond receive raw sewage they are called ponds (primary pond)
- Which of the following is not linked to biomagnification?
- Why frogs are helpful as bioindicators?
- For land base application soil type best suited to biotechnological intervention are (Sand and gravel)
- The septic tank can be a single-chamber tank or a two-compartment tank (called an Inhofe tank).
- In the Inhofe tank, settling occurs in the upper compartment (settling compartment).
- Attack by insect not only cause damage to the plant but also provide route?
- Insertion of foreign DNA in the vector resulting in the loss of the future (marker gene)
- Methogenic constortio of over 100 bacterial clone with the capability to digest (terephthalate)
- Preliminary treatment involves the removal of coarse solids only?

Due to continuous arrival of food in the form of BOD to the aeration tank, bacteria grow and reproduce continuously. If an indefinite population growth?

Were allow following problems raised:

The secondary sedimentation tank would become overloaded, the solids would not settle well and they would start to leave with the final effluent...

For higher efficiencies of phosphorus removal, effluent polishing methods can adopted?

Addition of coagulants
(Metallic ions):
Phosphorus precipitation
Effluent filtration: removal of the phosphorus present in the suspended solids in the effluent
Combination of the addition of coagulants and filtration

Wastewater is applied to the soil, supplying water and nutrients necessary for plant growth?

Some liquid evaporates, some percolates into the soil, and the largest fraction is absorbed by the plants...

Septic tanks are also a form of primary treatment?

The septic tanks and their variants, such as Inhofe tanks, are basically sedimentation tanks...

A number of conventional methods to remove Sculpture?

Methods for achieving a sculpture content reduction include washing pulverized coal and the use of fluidized bed technology, both organic and inorganic sculpture is present in coal, Biological methods: Aerobic, acidophilic chemolithotrophs like Thiobacillus species, for desulphurization of the inorganic sculpture in coal.

Bio filter consists of a relatively large vessel or container, typically made of cast concrete, metal or durable plastic?

The metals released in air by the processes of weathering, erosion and volcanic activity?

The nature of the soil or water which harbors the pollution can have a major effect on the actual expressed end-result?

Zooplanktons are microscopic animals living near to the surface of the water body. They are poor swimmers, instead relying on tides and currents as a transport mechanism. They feed upon phytoplanktons, bacterioplanktons, or detritus (i.e. marine snow)?

Phytoplankton's, also known as microalgae, are similar to terrestrial plants in that they contain chlorophyll and require daylight to live and develop

Mere a gene gene is one of a cluster of genes involved in bacterial detoxification of mercury, and is the one coding for the enzyme, mercuric ion reductase, which converts mercury from an ionic to a volatile form

The insect virus, Baculovirus, has been shown to be the method of choice for the over expression of genes in many applications of molecular biology

A strain of Pseudomonas putrid known to degrade PCBs

Facultative ponds which receive raw sewage is also known as primary ponds

Bio substitution are:

Biofuels,
Polymers,
Biodegradable

In the soil, a pollutant has basically four possible destinations:

- Retention in the soil matrix
- Retention by the plants
- Appearance in the underground water
- Collection by underdrains

Question and Answer

What are phase of bio filters?

The first methods to be developed. Consists of a relatively large vessel or container, typically made of cast concrete, metal or durable plastic, which holds a filter medium of organic material such as peat, heather, bark chips and the like. The gas to be treated is forced, or drawn, through the filter.

Three phase of bio filter?

This biofilter is a three-phase reactor composed of: Solid phase: consists of a support medium and biofilms, Liquid phase: consists of the liquid in permanent flow through the porous medium, Gas phase: formed by artificial aeration and by the gaseous by-products of the biological activity.

Importance of anaerobic zone?

The anaerobic zone gives good conditions for the development or selection of a large population of phosphorus accumulating organisms.

Transformation of plant resistance against herbicide?

The transfer of **herbicide resistance** genes from various sources into crop **plants** has been performed using several techniques. These transgenic products rely on both target site and metabolic detoxification **resistance** mechanisms.

Acidophile?

Acidophiles or **acidophilic** organisms are those that thrive under highly acidic conditions (usually at pH 2.0 or below). These organisms can be found in different branches of the tree of life, including Archaea, Bacteria, and Eukarya.

Phytoplankton?

Phytoplankton are microscopic marine algae. **Phytoplankton** is the base of several aquatic food webs. In a balanced ecosystem, they provide food for a wide range of sea creatures including whales, shrimp, snails, and jellyfish.

Function of bio trickling filter over bio filter?

A **biotrickling filter** is a combination of a **biofilter** and a bioscrubber. The bacteria responsible for decomposition are immobilised **on** a carrier or **filter** material. ... This water also carries away excess sludge/biofilm, as well as decomposition products which may hinder the biomass.

Role of phytoplankton for the transfer of pollutant among the different trophic level?

What is the role of phytoplankton in the transfer of pollutants among different trophic level?

Phytoplankton are the primary producers **in** aquatic ecosystems and play a key **role in** the **transport** of organic contaminants through the **food chain** to higher **trophic levels**. ... The bioaccumulation at lower **trophic levels** forms the first step **in** the **transfer** of chemicals through the **food chain**.

Advantage of bio indicator?

The most important reasons for using **bioindicators** are: the direct determination of **biological** effects, the determination of synergetic and antagonistic effects of multiple pollutants on an organism, the early recognition of pollutant damage to plants as well as toxic dangers to humans and.

Submerged bio filter?

Besides being a support medium for biomass growth, the granular material acts also as a filter medium. Periodic washings are necessary to eliminate the excess biomass accumulated.

Biotechnology and its all applications?

Biotechnology has **applications** in four major industrial areas, including health care (medical), crop production and agriculture, non-food (industrial) **uses** of crops and other products (e.g. biodegradable plastics, vegetable oil, biofuels, and environmental **uses**).

Pollutant?

A pollutant is a substance or energy introduced into the environment that has undesired effects, or adversely affects the usefulness of a resource.

Bio reactor?

a bioreactor is a vessel in which a chemical process is carried out which involves organisms or biochemically active substances derived from such organisms.

Glysohates?

Glyphosate is a broad-spectrum systemic herbicide and crop desiccant. It is an organophosphorus compound, specifically a phosphonate, which acts by inhibiting the plant enzyme 5-enolpyruvylshikimate-3-phosphate synthase. It is used to kill weeds, especially annual broadleaf weeds and grasses that compete with crops.

List two advantage of conventional activated sludge system?

High biomass concentration in the reactor due to the recirculation of the settled bacteria from the bottom of the secondary sedimentation tank.

Biomass remains in the system longer than the liquid, resulting in high BOD removal efficiency.

Environmental biotechnology?

Environmental biotechnology is biotechnology that is applied to and used to study the natural environment. Environmental biotechnology could also imply that one try to harness biological process for commercial uses and exploitation.

How can be prepared plant as recombinant?

In simpler terms, **recombinant DNA** is formed when the **DNA** from one organism is incorporated into the **DNA** of another organism of the same or different species. ... Using enzymes like ligase, the cut **DNA** fragment is attached to the vector. This is then introduced into the host cell, which takes it up as part of its own **DNA**.

How plants are modified as bio indicator?

Plant indicators The presence or absence of certain **plant** or other vegetative life in an ecosystem can provide important clues about the health of the environment: environmental preservation. There are several types of **plant** biomonitors, including mosses, lichens, tree bark, bark pockets, tree rings, and leaves.

What are thermophile? Where are they found?

A thermophile is an organism—a type of extremophile—that thrives at relatively high temperatures, between 41 and 122 °C. Many thermophiles are archaea. Thermophilic eubacteria are suggested to have been among the earliest bacteria.

Thermophiles are **found** in various geothermally heated regions of the Earth, such as hot springs like those in Yellowstone National Park (see image) and deep sea hydrothermal vents, as well as decaying plant matter, such as peat bogs and compost.

Three phase submerged aerated?

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Animal as bio indicator and give example?

A **bioindicator** is a living organism that **gives** us an idea of the health of an ecosystem. ... The numbers of earthworms in the soil can also be used to indicate the health of the soil. **One example** of a **bioindicator** is lichens.

What is the bio indicator?

Bioindicators are living organisms such as plants, planktons, animals, and microbes, which are utilized to screen the health of the natural ecosystem in the environment.

The expression 'Bioindicator' is used as an aggregate term referring to all sources of biotic and abiotic reactions to ecological changes.

They are used for assessing environmental health and biogeographic changes taking place in the environment e.g., a plankton responding rapidly to changes taking place in the surrounding

Importance of an aerobic zone?

It is essential to have anaerobic and aerobic zones in the treatment line for the biological removal of phosphorus. The anaerobic zone gives good conditions for the development or selection of a large population of phosphorus accumulating organisms.

Bio filter? What are the bio filter give example?

Biofiltration is a pollution control technique using a bioreactor containing living material to capture and biologically degrade pollutants.

Examples of biofiltration include: Bioswales, biostrips, biobags, bioscrubbers, Vermifilters and trickling **filters**. Constructed wetlands and natural wetlands. Slow sand **filters**.

Glyphosate?

'Glyphosate' (herbicide) is an analogue of phosphoenol pyruvate and inhibits the enzyme 5- enolpyruvylshikimate-3-phosphate synthase. The gene coding for this enzyme has been identified, isolated and inserted into a number of plants e.g., Petunias. The gene was expressed behind a CaMV promoter and introduced using *A. tumefaciens*.

How we improve crop quality?

Consider these 9 tips, techniques, and methods regarding how to increase your corn crop yield

1. Plant Early, Plant Effectively. ...
2. Practice Seasonal Soil Rotation. ...
3. Know The Yield Potential. ...
4. Always Scout Your Fields. ...
5. Ensure Proper Water Drainage. ...
6. Utilize Fertilizers. ...
7. Test Your Soil. ...
8. Weed Early and Often.

Two advantage of conventional system of sludge?

Advantages of Conventional Dispersed Plug Flow Process 1. Allows smaller volume than CSTR 2. Flexible operation, zone aeration, step feeding options, accommodates anoxic and aerobic processes with single biomass for biological nutrient removal 3. Less aeration than CSTR 4.

How marine plant indicate the health of environment give example?

Marine plants provide valuable information to predict the status of oceanic environment, as they are immobile and rapidly obtain equilibrium with their natural surrounding.

Changes in the diversity of species of phytoplankton, including *Euglena clastica*, *Phacus tortus*, and *Trachelon anas*, indicate the pollution of marine ecosystems

Long 5 Marks

Three phases of submerged aerated biofilter? Three phases of biofilm?

This biofilter is a three-phase reactor composed of:

Solid phase: consists of a support medium and biofilms,

Liquid phase: consists of the liquid in permanent flow through the porous medium,

Gas phase: formed by artificial aeration and by the gaseous by-products of the biological activity.

Genetic modification of popular plants how they remove mercury from plants?

Genetic modification of plants involves adding a specific stretch of DNA into the plant's genome, giving it new or different characteristics. This could include changing the way the plant grows, or making it resistant to a particular disease.

There are two types of **mercury removal** materials: non-regenerative sorbents and regenerative adsorbents. With non-regenerative types of **mercury** sorbents, the process fluid flows through the sorbent bed for a number of years, after which the sorbent is replaced.

Difference between transformation, transduction and conjugation?

In **transformation**, a bacterium takes up a piece of DNA floating in its environment. In **transduction**, DNA **is** accidentally moved from one bacterium to another by a virus. In **conjugation**, DNA **is** transferred **between** bacteria through a tube **between** cells.

Function of filter medium in bio filter and of compound used in?

The medium offers good water holding capacity and soluble chemicals within the waste gas dissolve into the film of moisture around the matrix. The medium itself provides physical support for microbial growth, with a large surface area to volume ratio, high in internal void spaces and rich in nutrients to stimulate and sustain bacterial activity

Recombinant microorganism particularly bacteria and yeast?

GE of microorganisms for use in environmental biotechnology: expansion of metabolic pathways either to modify the existent metabolic capability, or to introduce new pathways.

Many examples... A strain of *Escherichia coli* into which was engineered some 15 genes originating from *Pseudomonas*. These were introduced to construct a pathway able to produce indigo for the dyeing of denim (Bialy, 1997). The traditional method requires the use of toxic chemicals.